DIPLOMA PROGRAMME PRACTICE

Mathematics

Applications and interpretation SL

Set B / Paper 1 / Short response

90 minutes · 80 marks · All questions compulsory

Use a graphic display calculator and a clean Mathematics formula booklet.

Write your reasoning and working in the spaces provided; use additional answer paper if needed. Unless stated otherwise, give exact answers or round numerical results to three significant figures.

Diagrams are illustrative unless coordinates or dimensions are specified.

QUESTION 1

6 marks

A device stores 7.2×10^9 bytes of observations in a day. Each full archive block stores exactly 3.0×10^6 bytes. For this calculation take the quoted values as exact.

(a) Find the number of full blocks required for one day. Give your answer in ordinary notation. [2]

(b) Compression is expected to save 18% of the bytes, but the actual saving could be from 16% to 20%, inclusive. Find the smallest guaranteed sufficient number of blocks and explain your choice of percentage. [4]

QUESTION 2

6 marks

A battery delivers 84 km of range when new. At each annual inspection its measured range is modelled as 93% of the previous inspection's range.

(a) Write the model after n annual intervals and find the predicted range after three. [2]

(b) Replacement is required at the first inspection when range is below 55 km. Determine which inspection triggers replacement, justifying the integer choice. [4]

QUESTION 3

6 marks

From a lookout, an observer sights a buoy on a bearing of 040° and a second buoy on a bearing of 105°. The buoys are 2.8 km and 4.1 km from the lookout respectively.

(a) Find the included angle and the distance between the buoys. [3]

(b) A boat travels directly from the first buoy to the second at 12 km/h. Find the time in minutes. Explain why the result might underestimate the actual time. [3]

QUESTION 4

6 marks

Forty calls are grouped by duration t minutes. The frequencies for $0 \le t < 2$, $2 \le t < 4$, $4 \le t < 8$ and $8 \le t < 12$ are 6, 14, 12 and 8.

(a) Estimate the mean call duration. [3]

(b) What fraction of calls lasted at least 4 minutes? Explain why this is exact for the sample although your mean is an estimate. [3]

QUESTION 5

6 marks

The height of a launched water jet is $h(t) = 0.8 + 6t - 4.9t^2$ metres, where t is time in seconds after launch. The model ends when the jet first reaches the ground.

(a) Find the maximum height and the time at which it occurs. [3]

(b) Determine the model's time domain, and explain why the other algebraic zero is excluded. [3]

QUESTION 6

6 marks

A factory models fill mass X by a normal distribution of mean 506 g and standard deviation 3.8 g. Underweight packs have mass below 500 g.

(a) Find the probability that a randomly selected pack is underweight. [2]

(b) Find the 95th percentile of mass. A supervisor claims exactly 5 packs in every batch of 100 will exceed that mass. Comment. [4]

QUESTION 7

7 marks

Five film clips P, Q, R, S and T receive two sets of ranks (1 = best). The youth panel ranks them 1, 2, 3, 4, 5; the adult panel ranks them 2, 4, 1, 3, 5, in the same clip order.

(a) Calculate Spearman's rank correlation coefficient. [4]

(b) Explain why ranking observations can hide information. Does the coefficient show that both panels chose the same winner? [3]

QUESTION 8

7 marks

A cylindrical sample tube has internal radius 1.6 cm and a straight usable length of 11 cm. It may be filled to only 85% of its cylindrical volume. Treat the stated dimensions and fill fraction as exact in this model. Use 1 cm^3 = 1 mL.

(a) Determine its permitted liquid volume. [3]

(b) A technician must store 1.88 litres. Find the minimum number of tubes, and explain why rounding the capacity first to the nearest 10 mL could give a different decision. [4]

QUESTION 9

7 marks

Two mobile clinics are at U(−2, 1) and V(4, 1). All coordinates are in kilometres; travel is modelled along straight lines.

(a) Find the boundary between their nearest-clinic regions. Explain why it is vertical. [3]

(b) A river makes the actual journey from P(0, 3) to U twice the straight-line distance, while the journey to V equals its straight-line distance. Determine which clinic is now closer by journey distance. [4]

QUESTION 10

7 marks

At a fair a player pays 4 dollars and receives exactly one gross prize: 0 dollars with probability 0.60, 5 dollars with probability 0.30 or 20 dollars otherwise.

(a) Find the probability of the largest prize and the expected net gain to the player. [3]

(b) The fair keeps the fee and probabilities fixed. Find the largest prize that would make the game fair, and explain what fair means here. [4]

QUESTION 11

8 marks

The energy E in a small thermal store after t hours is modelled by $E(t) = 18 + 8t - t^2$ kWh, for $0 \le t \le 5$.

(a) Find E(2) and the slope of the tangent to the graph at t = 2. [3]

(b) Use the tangent to estimate E(2.5), and calculate the error in your estimate. [3]

(c) Explain geometrically why the same tangent also overestimates E(1.5). [2]

QUESTION 12

8 marks

Independent random samples of delivery times are taken for two firms. Assume approximately normal populations. A two-sided independent-samples t-test of equal population means returns p = 0.031.

(a) State both hypotheses in the context of this study. [2]

(b) State the decision and its contextual conclusion at 5%, then explain what changes at 1%. [4]

(c) Explain why the reported p-value alone cannot establish which firm is faster or whether the difference matters to customers. [2]